Yanbo Zhao

Raleigh, NC, USA yzhao62@ncsu.edu +1 (984)-289-5003

Education

North Carolina State University *Sept 2022 – Present*
Ph.D. in Computer Science, advised by Dr. Jiajia Li and Dr. Xu Liu *(Expected: May 2027)*
Shandong University *Sept 2019 – Jun 2022*
M.S., in Software Engineering, advised by Dr. Lei Ju.
Shandong University *Sept 2015 – Jun 2019*
B.S., in Software Engineering

Skills & Expertise

Programming Languages: C/C++, Python, Shell.
Parallel & Architecture: CUDA(PTX, SASS), OpenMP, Triton, ROCm, GPU Architecture.
Profiling Tools: Nsight Compute/System, NVBit, CUPTI, Compute Sanitizer API, HPCToolkit.
Systems & Performance: GPU/CPU co-scheduling, Memory redundancy detection.

Work Experience

Student Researcher, Google *May 2026 - Aug 2026*
Durham, NC, USA, Mentor: Dr. Xu Liu and Namhyung Kim

Research Intern, Pacific Northwest National Laboratory (PNNL) *Jun 2025 - Aug 2025*
Richland, WA, USA, Mentor: Dr. Ajay Panyala

Teaching/Research Assistant, North Carolina State University (NCSU) *Aug 2022 - Present*
Raleigh, NC, USA

Publications

TIDES: Tiered Block-Sparse Tensor Contraction with Streaming Transpose on GPUs *SC 26*
Zecheng Li, Sri H. R. Deverapalli, ***Yanbo Zhao****, Frank Mueller, Karl Pierce, Jiajia Li*

RedSan: Redundant Memory Instruction Sanitizer for GPU Programs *SC 25*
Yanbo Zhao*, Yueming Hao, Zecheng Li, Shuyin Jiao, Xu Liu, and Jiajia Li*

Leveraging AI Ecosystem for Portable and Sustainable GPU Kernels in HPC *ARRAY 26*
Yanbo Zhao*, Zhaonan Meng, Teja Manthena, Xu Liu, Ajay Panyala, Jiajia Li*

SparseZETA: Intelligent Auto-tuner for Designing High-Performance SpMV Programs *PLDI 26*
Zhen Du, Ying Liu, Xionghui Chen, ***Yanbo Zhao****, Xiaobing Feng, Huimin Cui, Jiajia Li*

cuThermo: Understanding GPU Memory Inefficiencies with Heat Map Profiling *arXiv Preprint*
Yanbo Zhao*, Jinku Cui, Zecheng Li, Shuyin Jiao, Xu Liu, and Jiajia Li*

Research Experience

cuVein: Understanding In-kernel Data Flow with Cross Grained Memory Centric Topology Distance Analysis (Under Review) @ NCSU *Jun 2025 - Present*

- Proposed Topology Distance, a metric linking a program's logical data flow to the physical communication costs of the underlying GPU hardware.
- Implemented cuVein to root-cause previously hidden performance bottlenecks in complex HPC kernels, providing data-driven guidance for optimizations not surfaced by official profilers.
- Enabled full-volume shadow memory analysis without sampling via a lock-free parallel analyzer and a contention-free shared-memory analyzer design, keeping end-to-end time overhead under 10x.

TypeCraft++: Fast, Zero-Shot Type Annotation for C++ Binaries (Ongoing) @ Google *May 2026 - Present*

- Designed a single-pass register type coloring algorithm that builds a PC-indexed register type table and

propagates types from DWARF-known PCs. Enables zero-shot annotation on binaries with only partial DWARF, achieving over 60% coverage on SPEC CPU2017.

- Replaced the prior per-instruction, forward-propagation-from-function-entry approach (informally "mock sample" annotation) with this single-pass algorithm, cutting annotation overhead by 5x and accelerating type annotation for PCs collected via live Linux perf sampling.

Leveraging AI Ecosystem for Portable and Sustainable GPU Kernels (ARRAY 2026) @ PNNL *Jun 2025 - Aug 2025*

- Developed an AOT compilation framework that integrates Triton with C/C++ HPC codebases, auto-generating optimized GPU code and a C++ wrapper into standalone shared libraries with the performance of hand-tuned CUDA.

RedSan: Redundant Memory Instruction Sanitizer for GPU Programs (SC 2025) @ NCSU *Dec 2024 - May 2025*

- A dynamic binary instrumentation tool that detects and quantifies redundant memory instructions in GPU applications without source code.
- Guided optimizations achieving up to 6.27× speedup and 3.00× fewer memory operations across HPC and AI benchmarks.
- Uncovered a previously undocumented compiler correctness issue with CUDA's ___*restrict*___ keyword and optimization opportunities within NVIDIA's cuSPARSE and cuSOLVER libraries.

cuThermo: Understanding GPU Memory Inefficiencies with Heat Map Profiling (Under Review) @ NCSU *Feb 2025 - Apr 2025*

- Developed cuThermo, a GPU memory profiler using dynamic binary instrumentation (NVBit) to identify runtime inefficiencies from compiled binaries without source code.
- Designed a heat map visualization based on a warp count metric, detecting five performance bottlenecks such as false sharing, memory misalignment, and shared memory abuse.
- Guided optimization of seven applications, achieving speedups of up to 721.79% on NVIDIA RTX 4090 (Ada) and A4500 (Ampere) GPUs.

Memogent: Exploring Reuse Distance Patterns in GPU Accelerated Applications @ NCSU *Jan 2023 - Jan 2024*

- Captured every GPU-kernel memory access using independent analysis kernels, identifying four reuse-distance inefficiency patterns; optimized 6 Rodinia/LULESH cases, cutting load memory traffic by 16.13% on average.

A profiling tool for cache persisting in NVIDIA GPU @ SDU *Jan 2021 - July 2022*

- Built an *Ampere* L2 persistence-cache tool that uses reuse-distance and memory-utilization analysis to suggest data to lock into L2, improving L2 hit rate and execution time.

Low Power Object Detection Challenge (DAC-SDC 2018) (Collaborated) @ SDU *Dec 2017 - May 2018*

- Optimized a darknet YOLOv2 detector for NVIDIA Jetson TX2 via multi-threaded pre-processing, pruning, and quantization, reaching 68.4% IoU at 23.64 FPS — 3rd place worldwide in the GPU category.

Service

SC 26 — Int'l Conf. for High Performance Computing, Networking, Storage, and Analysis — *Program Committee, Student Volunteer*
ICS 27 — Int'l Conf. on Supercomputing — *Program Committee*
ISC 26 — ISC High Performance — *Program Committee*
IEEE BigData 26 — IEEE Int'l Conf. on Big Data — *Program Committee*
SPAA 25 — Symp. on Parallelism in Algorithms and Architectures — *Junior Program Committee*
PPoPP 25 — Symp. on Principles and Practice of Parallel Programming — *Artifact Evaluation Committee*